EOF and SVD

假设我们原始数据距平为$X_{m\times n}$, 有m个空间点和n个时间点，数学意义上，即 X 的数据被记录在一共 m 个维度上，在第 i 维度的第 j 时刻，数据为$x_{i,j}$. 而EOF的本质就是将 X 重新分配到新的维度上：

$$X_{m\times n} = EOF_{m\times m} \cdot PC_{m\times n} = [\,eof_1, eof_2, \ldots, eof_m\,]_{m\times m} \cdot \begin{bmatrix} pc_1 \\ pc_2 \\ \ldots \\ pc_m \end{bmatrix}_{m\times n}$$

where EOF is the eigenvector matrixs, which is an normal orthogonal matrix, i.e.: $EOF^T = EOF^{-1}$,and PC is the projection of X in the new coordinates EOF:

$$PC = EOF^{-1} \cdot X = EOF^T \cdot X$$

Eigenvalues method

To calculate the eigenvmatrix EOF, first we should calculate the Covariance matrix:

$$C_{m\times m} = \frac{1}{n} X \cdot X^T$$

then the eigenvector matrix EOF or V satisfies:

$$V \cdot C = \Sigma \cdot C \tag{1}$$

where Σ is the eigenvalues matrix:

$$\Sigma = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \ldots & \lambda_m \end{bmatrix}$$

To solve (1), move the right hand side of the equation to lhs, get:

$$(V - \Sigma) \cdot C = 0 \tag{2}$$

now the equation becomes an first order homogeneous equation. To get a nontrivial solution for (2), the determinant of $(V - \Sigma)$ should be equals 0:

$$|V - \Sigma| = 0$$

from which we can calculate the V and Σ

SVD method:

$$X_{m,n} = svd(X) = U_{m,m}\Sigma_{m,n}V_{n,n}$$

where $U_{m,m} = EOF_{m,m}$, $PC_{m,n} = \Sigma_{m,n}V_{n,n}$.

Relationship of difference eigenvectors (eof_i)

To find out the relationship of difference eof_i, we can calculate:

$$\begin{aligned}
EOF^T \cdot EOF &= \begin{bmatrix} eof_1 \\ eof_2 \\ \dots \\ eof_m \end{bmatrix}_{m,1} \cdot \left[eof_1, eof_2, \dots, eof_m \right]_{1\times m} \\
&= \begin{bmatrix} eof_1 \cdot eof_1,\ eof_1 \cdot eof_2, \dots, eof_1 \cdot eof_m \\ eof_2 \cdot eof_1,\ eof_2 \cdot eof_2, \dots, eof_2 \cdot eof_m \\ \dots \\ eof_m \cdot eof_1,\ eof_m \cdot eof_2, \dots, eof_m \cdot eof_m \end{bmatrix}_{m\times m} \\
&= \begin{bmatrix} 1 & 0 & 0\dots & 0 \\ 0 & 1 & 0\dots & 0 \\ 0 & 0 & 1\dots & 0 \\ 0 & 0 & 0\dots & 1 \end{bmatrix}
\end{aligned}$$

or in other form, using the normal orthogonal matrix U:

$$EOF^T \cdot EOF = U^T \cdot U = I$$

that is, eigenvectors are independent from each others.

Relationship of difference PCs

$PC = [pc_1, pc_2, \dots, pc_m]^T = \Sigma V$, Similarly,

$$\begin{aligned}
PC \cdot PC^T &= \begin{bmatrix} pc_1 \cdot pc_1,\ pc_1 \cdot pc_2, \dots, pc_1 \cdot pc_m \\ pc_2 \cdot pc_1,\ pc_2 \cdot pc_2, \dots, pc_2 \cdot pc_m \\ \dots \\ pc_m \cdot pc_1,\ pc_m \cdot pc_2, \dots, pc_m \cdot pc_m \end{bmatrix}_{m\times m} \\
&= \Sigma V \cdot (\Sigma V)^T \\
&= \Sigma V V^T \Sigma^T \\
&= \Sigma^2
\end{aligned}$$